

QMS SOP TEMPLATE:

Out-of-Specification

Quality
Management
System
Standard
Operating
Procedure



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Out-of-Specification

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Introduction

This SOP is intended to prevent the discarding of analysis results that are out-of-specification, without a detailed and well documented investigation into the root cause of this aberrant result. Finding the root cause does not only extend the investigation to the laboratory but can also encompass the production processes.

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1.0 Purpose

This procedure outlines the steps to be followed when an OOS result is identified during testing in a laboratory or manufacturing environment. It ensures a systematic approach to investigate, document, and resolve OOS findings.

2.0 Scope

This procedure applies to all laboratory or manufacturing activities that involve testing and analysis, including raw materials, in-process materials, finished products, stability testing, and validation activities. This procedure does not apply to excursions of Environmental Monitoring Data.

3.0 Definitions/Abbreviations

Quality Assurance/Qualified Person: A designated individual responsible for ensuring that each batch of a medicinal product is manufactured and tested in accordance with applicable regulations before it is released for sale or supply.

GMP: A set of guidelines and regulations that ensure that pharmaceutical products are consistently produced and controlled according to quality standards. These practices are designed to minimize risks involved in the production process, ensuring that products are safe, effective, and of high quality.

Quality Control (QC): A systematic process of testing and inspecting raw materials, in-process materials, and finished products to ensure they meet predefined quality standards and regulatory requirements.

4.0 Responsibilities

Quality Control (QC) Analyst: identify and report OOS results.

Laboratory Manager: Review and approve the OOS report.

Investigation Team: Perform root cause analysis and determine corrective actions

Quality Assurance/Qualified Person: Ensure that the investigation is carried out following the procedure and document results, make final decision on product quality.

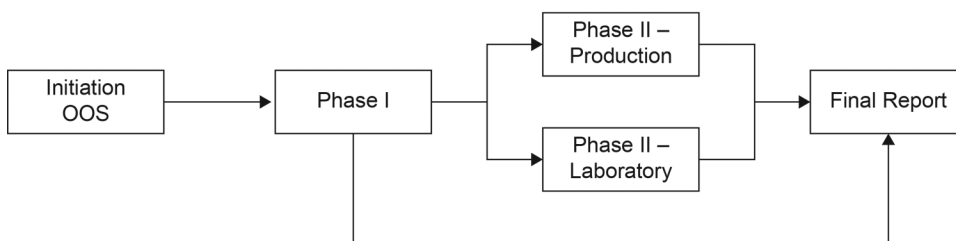
OOS Coordinator: Administrates the process, tracks timelines, organizes meetings on investigations, summarizes information.

Management: Approve conclusions and necessary corrective actions.

5.0 Requirements

- OOS Form (hard copy or electronic system generated)
- OOS Log (hard copy or electronic system generated)

6.0 Summary/Flowchart



7.0 Procedure

7.1 Identification of OOS Result

- Upon receiving test results that fall outside the established specifications, the analyst must immediately report the OOS result to the Laboratory Supervisor or Manager. The analyst must secure, if possible, the reagents and solutions used, to be able to use these for further investigation.
- The OOS result should be recorded in the appropriate testing log, and an OOS investigation form should be initiated.

7.2 Preliminary Review (Phase I)

An initial review is performed to verify and document that no mistakes were made during analysis.

The following activities are part of this Phase I:

- Discussion of analytical method followed to verify the correct method is used and that the analyst is sufficiently knowledgeable of the method.
- Verification of training and qualification records of the analyst
- Investigation into the raw data, including graphs and/or spectra, to abnormal or suspicious data.
- Verification of calculations used to determine the final result for correctness, applicability and scientific soundness
- Verification of GMP status/calibration of instruments used for testing
- Verification of standards, solutions and reagents used for testing
- Verification of internal standards and/or system suitability and comparison with validation and historical data
- Creation of hypothesis on possible errors that might have occurred and possible investigations into these hypotheses
- Documentation of the investigation results

7.3 Retesting (if applicable)

- If the OOS result is identified on a single test, perform a second test on the same sample to verify the result. Ensure that the sample is handled according to the same procedure as the original test.

- If the retest result is within specification, document the findings and conclude the investigation as “no issue identified.”
- If the retest result is also out-of-specification, proceed to Phase III.

7.4 Investigation and Root Cause Analysis (Phase II)

If production occurred by a third party, this party is involved in the investigation.

The investigation is performed by the OOS Coordinator and involves Production as well as any other department that might be involved or able to contribute.

During the investigation the following activities are performed:

- Review of Production Batch Documentation
- Review of raw material batch and lot number
- Review of all equipment used (including calibration records)
- Interviews with personnel involved
- Review of testing methodology
- Review of associated Deviations and Change Controls

Documentation of this phase includes:

- A clear description of the reason for the investigation
- A summary of aspects of the Production process that might have contributed to the out-of-specification
- Results from the review of production related documentation with identification of most likely root cause (s)
- A description of all actions taken

When other batches might also be affected, a Deviation is initiated.

When it is confirmed that an error was made during production, the OOS can be closed and the batch can be rejected. If this is not the case, Phase III of the Laboratory

Investigation will be started.

7.5 Phase III – Laboratory Investigation

This investigation is performed by the Quality Control (QC) Analyst together with the Investigation Team.

During the Laboratory Investigation the following activities can be performed:

- Expected root cause in the analysis: Retesting of the original, homogenous sample, that led to the OOS in the first place. The retest is performed by another analyst that has at least the same qualification as the analyst that got the OOS result. The hypothesis is documented upfront, including the number of retests to be performed.
- Expected root cause in sampling: Resampling the batch according to normal procedure, followed by testing.

When this deeper Laboratory Investigation reveals a clear mistake in analysis or sampling, the first obtained OOS result can be invalidated and replaced with the new result.

If no clear mistake in sampling or testing can be found, all results are documented and taken into account during appraisal of the batch.

- The investigation team should assess possible reasons for the OOS result. The root cause can be one of the following:
 - Instrumentation or equipment malfunction
 - Operator error
 - Contaminated or incorrect sample
 - Issues with raw materials
 - Issues with test methods or procedures

7.6 Final Report

The final report encompasses all steps taken and all (interim) conclusions, reporting of the end result(s), if any root cause could have been determined and the final appraisal of the batch.

Also any follow-up activities agreed, either short or long-term and summarized.

8.0 Calculations

Calculations on percentage of initials and confirmed OOS results per total of generated results gives an indication of performance, related to these aberrant results.

9.0 Documentation of Results

All OOS results are documented using the OOS form.

10.0 References

- 21 CFR Part 210 – Current Good Manufacturing Practice in Manufacturing, Processing, Packing, or Holding of Drugs; General
- 21 CFR Part 211 – Current Good Manufacturing Practice for Finished Pharmaceuticals
- EU GMP Guide – Part I: Chapter 6 (Quality Control)
- Pharmaceutical Inspection Co-operation Scheme. *PIC/S Guide to Good Manufacturing Practice for Medicinal Products: Part I (PE 009-17)*
- TRS 986 – Annex 2: WHO Good Manufacturing Practices for Pharmaceutical Products: Main Principles
- ICH Q10 – Pharmaceutical Quality System

- United States Pharmacopeial Convention. USP General Chapter <1058> Analytical Instrument Qualification
- Investigating Out-of-Specification (OOS) Test Results for Pharmaceutical Production, Guidance for Industry, CDER May 2022, Rev. 01

11.0 Revision History

- Rev 01 – Initial release
- Rev 02 – [To be updated with future changes]

12.0 Addenda

12.1 Addendum I: Out-of-Specification (OOS) Form Examples

Example: Phase I Laboratory Investigation

OOS number: OOS-24003	Analytical Method number: QC-778 pH testing
Product involved: Stability Sample 24 months SPIE capsules	Batch Number: ABC 33425
Result: pH 6.9	Specification: 7.0 – 9.0
Name Analyst: PEPA	Date of analysis: 19 NOV 2024
1. The analytical method used was discussed with the analyst and the used method was correct and correctly executed. The analyst was sufficiently trained and qualified. Please add (or refer to) supporting data. Yes / No Comments: Training records PEPA are attached.	
2. The raw data, including graphs and/or spectra, are reviewed and no suspicious or abnormal data were detected. Yes / No Comments: All documentation is checked and no corrections or changes were made. Also the print out from the pH meter is attached.	
3. The calculations used are verified for correctness, applicability and scientific soundness. No mistakes were found. Yes / No Comments: Calculations are not applicable.	

Example: Phase I Laboratory Investigation (continued)

OOS number: <i>OOS-24003</i>		Phase I Laboratory Investigation
4. <i>The GMP status of the used instruments is correct and they demonstrate to function properly.</i> Yes / No Comments: <i>pH meter is maintained, and daily calibrated; no abnormalities in the logbook. The electrode is clean and shows no signs of damage.</i>		
5. <i>All the standards, reagents and solutions have been verified and are according specifications.</i> Yes / No Comments: <i>The calibration buffers were measured against a new set of calibration buffers with correct results.</i>		
6. <i>All internal method verifications (internal standards, system suitability) are comparable to historical data and the validation data.</i> Yes / No Comments: <i>pH for previous time points of this batch were also on the low side of the specification (t = 6 months: pH = 7.3, t = 12 months: pH = 7.2, t = 18 months: pH = 7.0).</i>		
7. <i>Hypothesis have been formulated on possible errors or mishaps during the analysis.</i> Yes / No / NA If Yes, what are the hypothesis and how have these been verified? What was the result? Please add associated documentation. Comments: <i>Measuring a pH is a very simple method and no dilution for this sample was required.</i>		
Conclusion: <i>This is a true OOS result for pH. An investigation into similar batches of product should take place and a possible recall should be considered.</i>		
Performed by:	Name and Signature: <i>Shawn Price</i> SPrice	Date: <i>19 NOV 2024</i>
Approved by:	Name and Signature: <i>James Jones</i> JJones	Date: <i>19 NOV 2024</i>

Example: Phase II Production Investigation

OOS Number:		Phase II Production Investigation	
Related Batch Number			
1. Reason for investigation/hypothesis			
2. Summary of aspects of production process that might have created the OOS			
3. Results of review of production related documentation and (probable) cause			
4. Results of assessment of previous occurrence of the problem, including reference batches that might have a similar issue:			
5. List of initiated CAPA's:			
6. Is further laboratory investigation required?			
Yes / No			
Comment:			
Performed by:	Name and Signature:	Date:	
Approved by:	Name and Signature:	Date:	

Example: Phase II Laboratory Investigation

OOS Number:		Phase II Laboratory Investigation	
Related Batch Number			
1. Expected cause in analysis?			
Yes / No If Yes, please add data of investigation and conclusions.			
2. Expected cause in sampling?			
Yes / No If Yes, please add data of investigation and conclusions.			
3. List of initiated CAPAs			
Final conclusion:			
Performed by:	Name and Signature:	Date:	
Approved by:	Name and Signature:	Date:	



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